

GateWatch

How the Meshtastic Fence Monitoring System Works

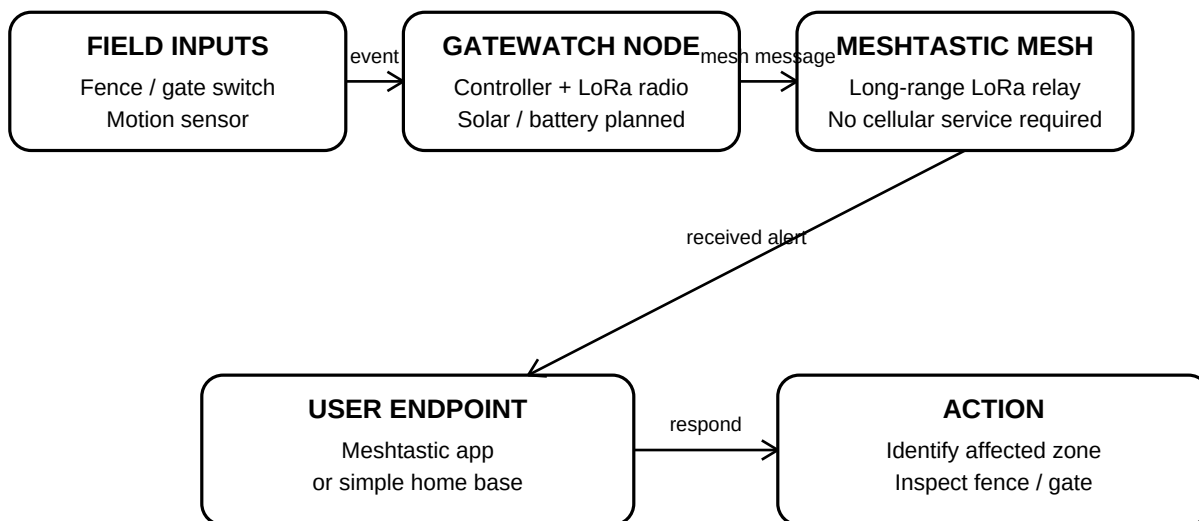
Project: GateWatch / Farmer's Fence Mesh	Build-Off: Meshtastic Build-Off 2026
Purpose: Remote fence, gate, and movement alerts	Status: Concept / design; physical validation pending

1. What GateWatch Is

GateWatch is a proposed low-power monitoring system for farms and ranches where long fence lines may be outside cellular or Wi-Fi coverage. A small field node watches for conditions such as a fence breach, an open gate, or nearby movement. When an event is detected, the node sends a message through a Meshtastic LoRa mesh to the farmer's phone or a simple home-base device.

The design is intended to reduce the time spent manually checking remote fence sections and to provide an early warning when livestock containment or property security may be affected.

2. System Architecture



The key idea is simple: sensors generate an event, a field node converts that event into a message, Meshtastic carries the message across the property, and the user receives enough information to decide where to inspect.

3. How an Alert Moves Through the System

1. Monitor

A GateWatch field node continuously watches connected inputs. Planned examples include a fence or gate switch and a motion sensor.

2. Detect

If an input changes state - for example, a gate opens or a fence-monitoring switch indicates a possible breach - the controller records the event.

3. Package

The controller creates a short alert message identifying the node or monitored zone and the type of event.

4. Transmit

The LoRa radio sends the alert into the Meshtastic network. Nearby Meshtastic nodes can relay the packet, allowing coverage beyond a single direct radio link.

5. Receive

The alert reaches a Meshtastic-connected phone or a simple wireless home-base device.

6. Respond

The user can identify the affected section and inspect the fence, gate, post, or surrounding area.

4. Main System Blocks

Block	Role in GateWatch
Field sensors	Detect physical changes such as gate state, fence condition, or motion.
Controller	Reads sensors, applies event logic, and creates an alert.
LoRa radio	Provides long-range, low-bandwidth wireless communication.
Meshtastic mesh	Relays messages between nodes without requiring normal cellular infrastructure.
Power system	Planned solar and battery power for remote, low-maintenance operation.
User endpoint	Meshtastic app or simple base station used to receive alerts and identify the affected area.

5. Example Deployment

A ranch could place GateWatch nodes at high-risk locations such as gates, isolated fence corners, road crossings, or sections frequently damaged by wildlife or fallen branches. The nodes would communicate through the Meshtastic mesh, so a node that cannot directly reach the house could potentially pass its alert through another mesh node.

Example event:

Fence section B changes state -> GateWatch Node B detects the event -> the node transmits an alert -> the message is relayed through the Meshtastic mesh -> the farmer receives the alert and checks section B.

6. Why Meshtastic Fits the Concept

GateWatch is aimed at locations where conventional infrastructure may be unavailable or undesirable. Meshtastic uses LoRa radios and mesh networking, making it suitable for small event messages over comparatively long distances while allowing intermediate nodes to relay traffic.

The project is not intended to stream video or large amounts of data. Its job is to move small, useful status and alarm messages from remote points back to the user.

7. Current Project Status

As of the Meshtastic Build-Off 2026 final review period, GateWatch should be treated as a concept and system design rather than a completed, field-validated product. The physical prototype, measured radio range, solar/battery endurance, environmental enclosure performance, and real-world sensor behavior still require testing.

This document describes the intended operation of the system. It does not claim that the full system has already been assembled or validated in the field.

8. Next Build Milestones

- Assemble one field node and verify its sensor inputs.
- Establish Meshtastic radio communication between the field node and a receiving node.
- Generate real fence/gate and movement events and confirm alert delivery.
- Measure practical range on representative rural terrain.
- Measure current draw and validate the solar/battery power budget.
- Install the electronics in a weather-resistant enclosure and perform an outdoor field trial.

Repository

<https://github.com/Shanker511/GateWatch>

Prepared as a concise technical overview for the GateWatch Meshtastic Build-Off project.